

2 All the marks on the mark scheme are designed to be awarded; mark schemes should be applied positively. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme. If there is a wrong answer (or no answer) indicated on the answer line always check the working in the body of the script (and on any diagrams), and award any marks appropriate from the mark scheme.

Questions where working is not required: In general, the correct answer should be given full marks.
Questions that specifically require working: In general, candidates who do not show working on this type of question will get no marks – full details will be given in the mark scheme for each individual question.

9 Linear equations
Unless indicated otherwise in the mark scheme, full marks can be gained if the solution alone is given on the answer line, or otherwise unambiguously identified in working (without contradiction elsewhere). Where the correct solution only is shown substituted, but not identified as the solution, the accuracy mark is lost but any method marks can be awarded (embedded answers).

Guidance on the use of abbreviations within this mark scheme	
M	method mark awarded for a correct method or partial method
P	process mark awarded for a correct process as part of a problem solving question
A	accuracy mark (awarded after a correct method or process; if no method or process is seen then full marks for the question are implied but see individual mark schemes for more details)
C	communication mark awarded for a fully correct statement(s) with no contradiction or ambiguity
B	unconditional accuracy mark (no method needed)
oe	or equivalent
cao	correct answer only

17	4 : 1	P1	for associating algebraic expressions with the correct ratio eg $p - 5 : q - 5$ ($= 5 : 1$) or $p + 20 : q + 20$ ($= 5 : 2$)	Award for one of the two simultaneous equations eg $5q - p = 20$, $5q - 2p = -60$ oe Award for a simultaneous equation method to eliminate one variable leading to either $p = 80$ or $q = 20$ Award for a simultaneous equation method to eliminate both variables leading to either $p = 80$ and $q = 20$
		P1	for $\frac{p + 20}{q + 20} = \frac{5}{2}$ or $\frac{p - 5}{q - 5} = \frac{5}{1}$ oe or $p - 5 = 5(q - 5)$ or $2(p + 20) = 5(q + 20)$ oe	
		M1	for a complete method shown to find p or q	
		M1	for a complete method shown to find p and q or two values for p and q that are in the ratio 4 : 1 or an unsimplified ratio 4 : 1 (eg 80 : 20) or an answer of 1 : 4	
		A1	cao	